

Technology Stack

Date	29 June 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Technology Stack:

identify the programming languages, frameworks, databases, front-end technologies, back-end technologies, and APIs that will be used to build our solution. A well-chosen technology stack ensures the application is scalable, maintainable, secure, and meets the project's functional and technical requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, Bootstrap 5, Jinja2 Templates	HTML/CSS with Bootstrap 5 provides a responsive, mobile-friendly UI. Jinja2 templates integrate seamlessly with Flask, enabling dynamic rendering of approval/rejection results without a separate front-end framework.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python 3.10, Flask Framework	Python is the industry standard for machine learning development. Flask is lightweight and easy to integrate with scikit-learn and XGBoost models, enabling rapid REST API creation for real-time credit card approval predictions.
3	Machine Learning Libraries (e.g., scikit-learn, TensorFlow, Keras)	scikit-learn, XGBoost, Pandas, NumPy	scikit-learn provides Logistic Regression, Random Forest, and Decision Tree classifiers. XGBoost delivers high accuracy on tabular financial data. Pandas and NumPy handle feature engineering and data preprocessing pipelines.
4	Model Persistence & Serialisation (e.g., Pickle, Joblib, ONNX)	Pickle / Joblib	The best-performing ML model is serialised using Pickle/Joblib so it can be loaded directly into the Flask application at runtime without retraining, ensuring fast and consistent predictions.

5	Cloud / Hosting / Deployment (e.g., AWS, Azure, IBM Cloud, Heroku)	IBM Watson Machine Learning (IBM Cloud)	IBM Watson Machine Learning provides a managed cloud deployment pipeline for the trained model. It enables scalable, real-time REST API endpoints for credit card approval predictions, accessible from any client application.
6	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub	GitHub is used for source code versioning and team collaboration. It tracks all changes to the ML pipeline, Flask application, and deployment scripts, ensuring reproducibility and enabling rollback if needed.
7	Data Storage / Dataset Management (e.g., CSV, MySQL, MongoDB)	CSV Files (UCI Credit Card Dataset)	The UCI Credit Card Dataset (CSV format) is used for model training and evaluation. CSV storage is lightweight and directly compatible with Pandas for preprocessing. No live database is required as decisions are made on-the-fly.
8	Third-Party APIs / Other Tools (e.g., IBM Watson, Postman, Jupyter)	IBM Watson ML API, Jupyter Notebook, Postman	IBM Watson ML REST API exposes the deployed model for external consumption. Jupyter Notebook is used for EDA, model training, and evaluation. Postman validates API endpoints before Flask integration